2026 MIDDLE SCHOOL STEM CAMPS

PY-STEM

INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

IN THIS PACKET

INSTRUCTOR · PYS-01 · BIOMEDICAL

Magnetic Capsule Maze Cup

Steer a pill-camera without touching it

At a glance: Steer a magnetic pill-camera through a body maze without touching it. About 80 min; 4 to 6 teams.

MATERIALS

- 6 mm driver magnets in bottle caps; keep the tiny 2 by 1 mm magnets locked away
- maze boards
- steel washers, nuts, or paper clips
- timers
- barrier sheets

FACILITATION ARC

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- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does moving the magnet slowly help?
- How does distance change the pull you feel?
- Where is remote actuation used in real medicine?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Fastest clean run	50
Low penalties	20
Magnetic-control explanation	20
Redesign	10
Total	**100**

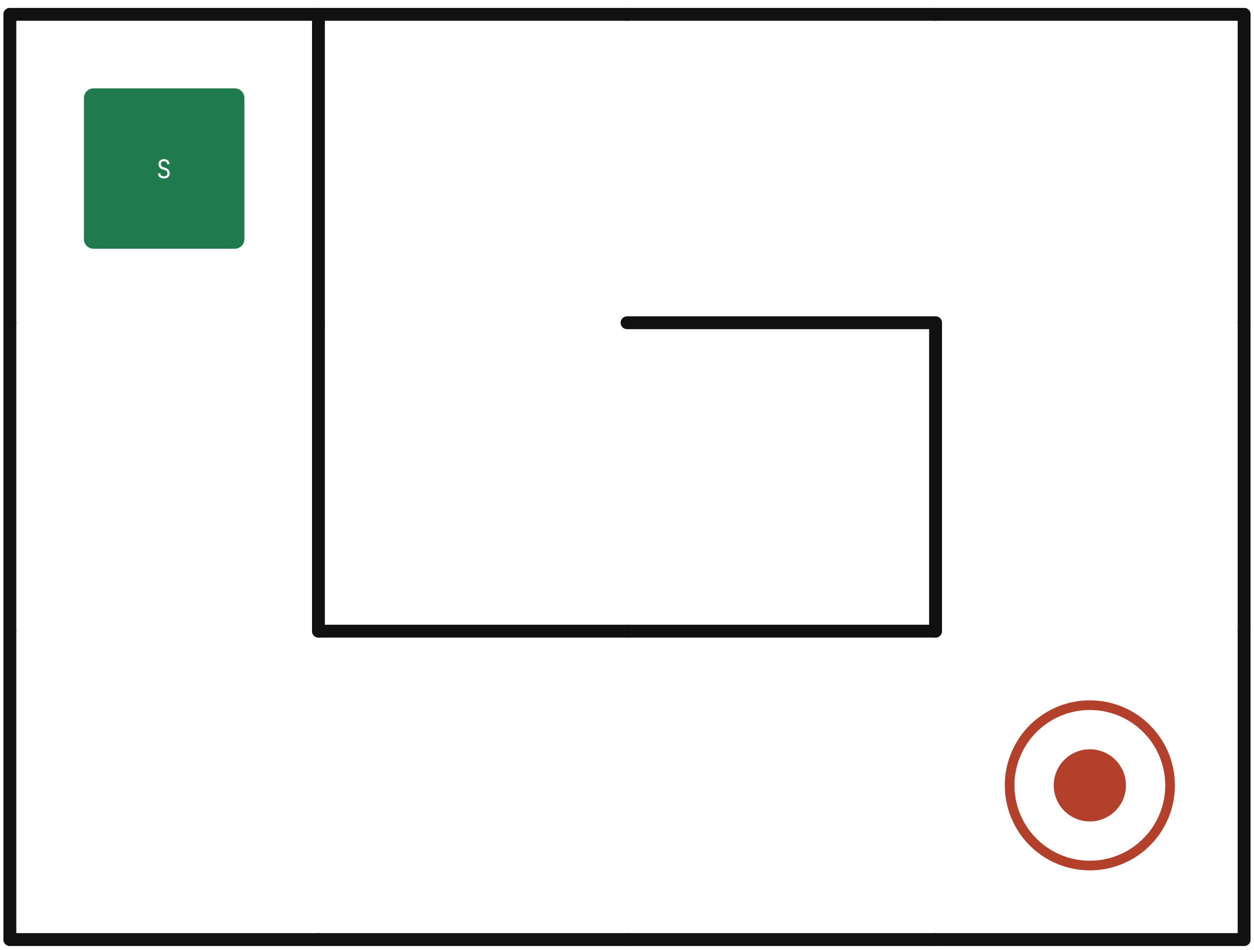
SOURCES Science Museum Group Magnetic Maze

Maze boards (laminate; one board per team)

Print on cardstock and laminate, one board per team at the difficulty you assign. The green square **S** is the start; the red target is the finish. Place a steel washer or small nut (the "capsule") on S and move the 6 mm driver magnet UNDER the board to drag it to the target without touching the maze walls. A small washer turns corners more easily than a long paperclip, especially on Maze D. Move the driver slowly, and pre-test each token through the laminated sheet before the run.

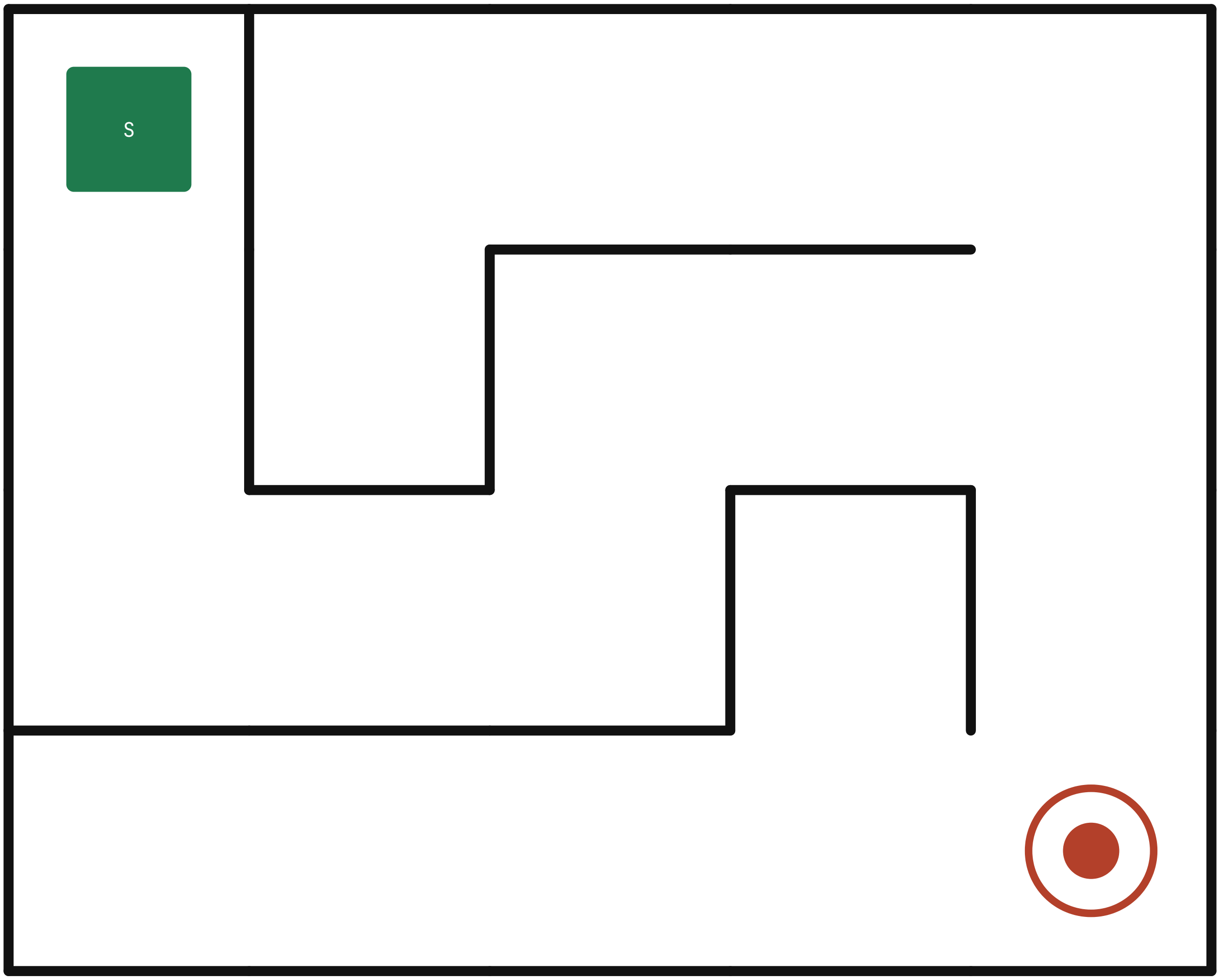
Maze A

WARM-UP



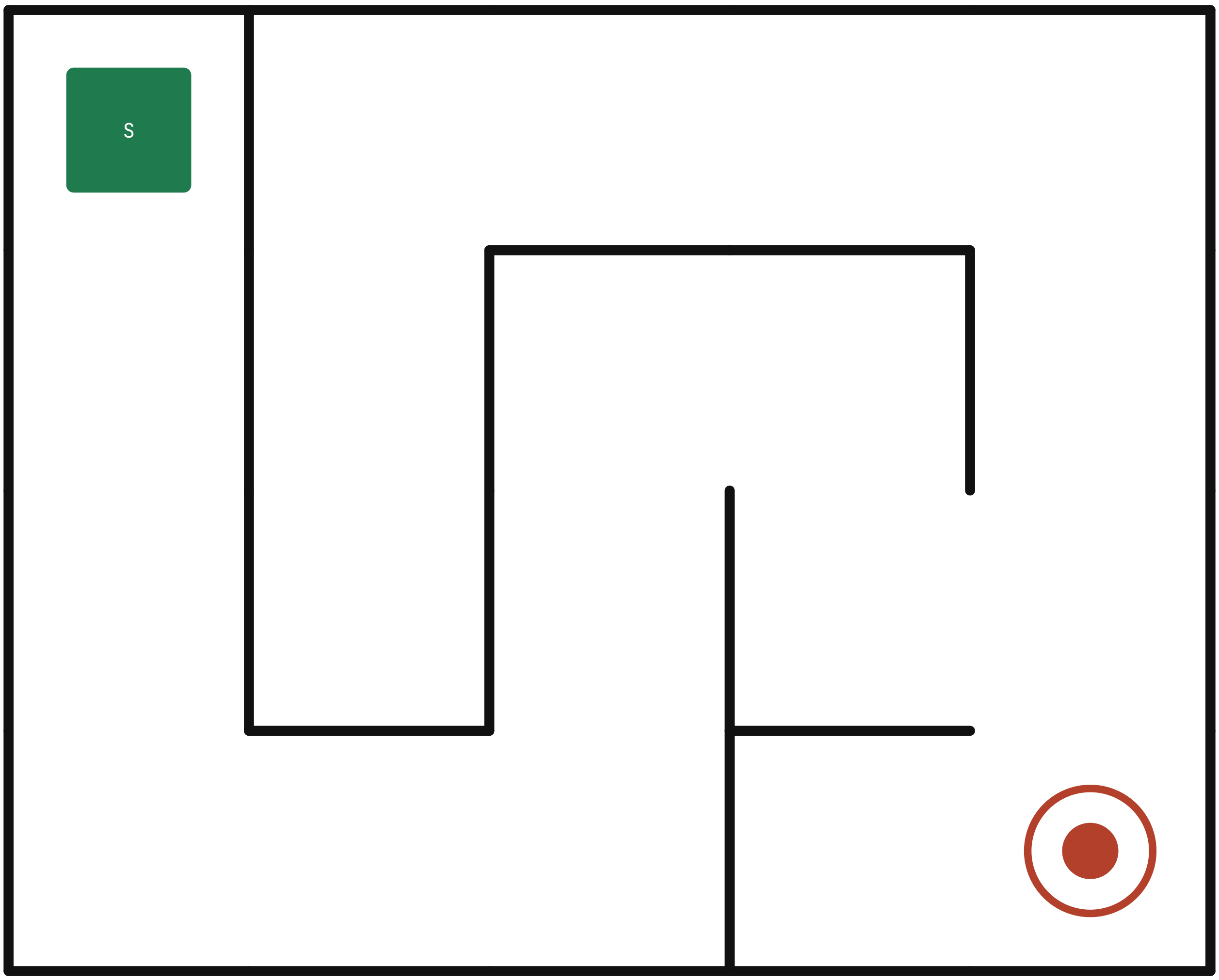
Maze B

EASY



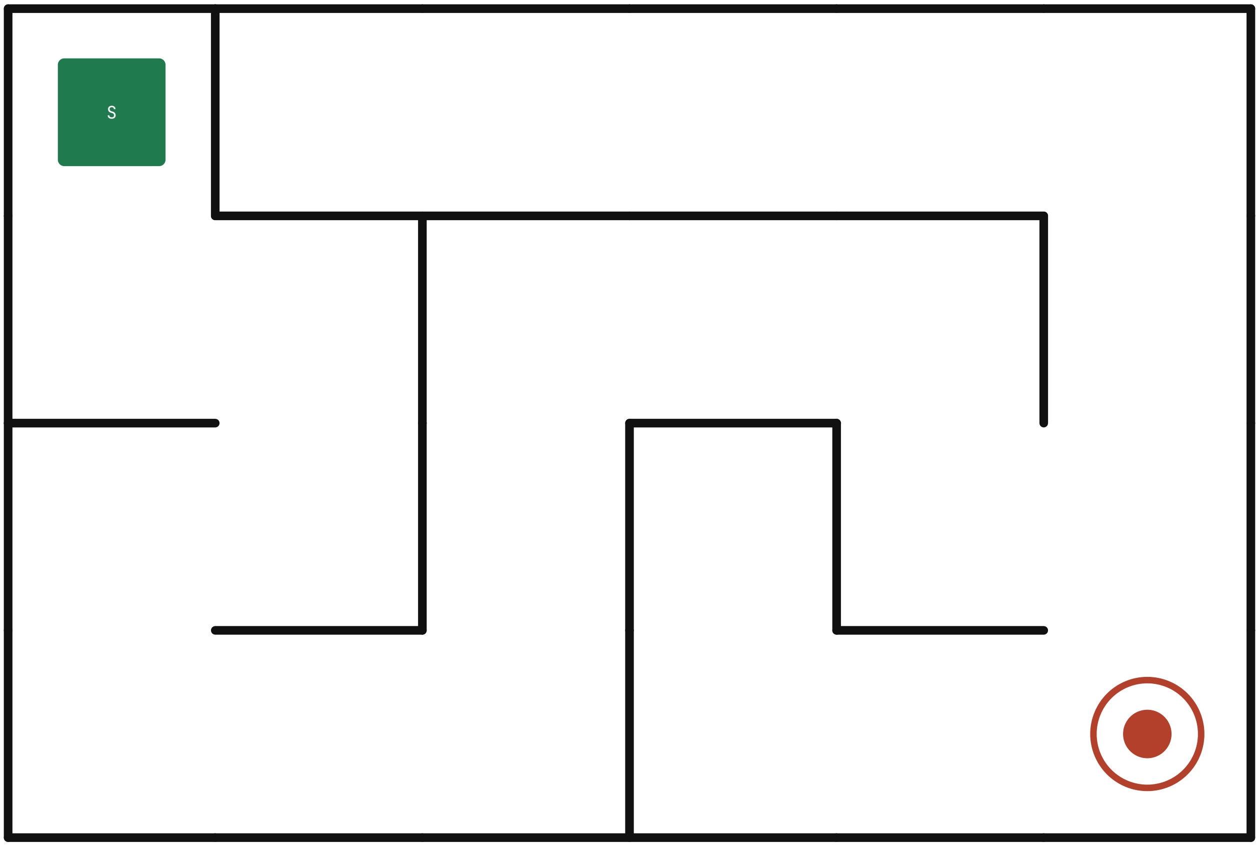
Maze C

MEDIUM



Maze D

HARD



Oobleck Armor Arena

Armor from a liquid that hardens on impact

At a glance: Design armor with a material that acts like a liquid until impact. About 80 min; 4 to 6 teams.

MATERIALS

- cornstarch
- water
- zip bags
- cardboard sleeves, a target (coin stack or foam square), and a rigid lid for the press
- test weights, and a sealed corn-allergy pad (teacher-made)
- spoons
- trays

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DEBRIEF

- Why does a fast press feel harder than a slow one?
- How little oobleck still protected the target?
- Where would shear-thickening armor be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Protection result	40
Material efficiency	20
Explanation	20
Redesign	10
Cleanup	10
Total	**100**

SOURCES Science Buddies Oobleck; Exploratorium Ooze

Cardboard Automata Arcade

One crank, one clever motion

At a glance: Build a one-crank machine that performs a mini task. About 80 min; 4 to 6 teams.

MATERIALS

- cardboard
- skewers
- straws
- craft sticks
- tape
- hot glue station
- scissors

FACILITATION ARC

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DEBRIEF

- How did your cam shape change the motion?
- What caused jams, and how did you fix them?
- Where do cams show up in real machines?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Reliable motion	30
Cam or linkage complexity	20
Task success	20
Explanation	20
Aesthetics and build quality	10
Total	**100**

SOURCES Exploratorium Cardboard Automata

Stethoscope Sprint and Recovery

Build a listening tool, measure like a scientist

At a glance: Build a medical listening tool and use it like a sports scientist. About 80 min; 4 to 6 teams.

MATERIALS

- funnels
- flexible tubing
- tape
- balloons or membranes
- stopwatches
- sanitizing wipes

FACILITATION ARC

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DEBRIEF

- What change most improved your sound quality?
- Why does a good seal matter so much?
- What does a fast recovery suggest?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Sound quality	30
Measurement accuracy	25
Explanation	20
Redesign	15
Data clarity	10
Total	**100**

SOURCES Science Buddies Make a Stethoscope

Reaction Time Combine

Prove which strategy cuts your delay

At a glance: Train like a sports scientist and prove which strategy cuts delay. About 80 min; 4 to 6 teams.

MATERIALS

- meter sticks
- reaction-time strip (this packet)
- pencils
- score sheets
- clipboards

FACILITATION ARC

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DEBRIEF

- Why use the median instead of your fastest catch?
- Which strategy helped, and why might it?
- Where does reaction time matter in real life?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Best median reaction time	35
Data quality	20
Strategy improvement	20
Explanation	15
Teamwork	10
Total	**100**

SOURCES Science Buddies Reaction Time

SONAR Slinky Showdown

Make invisible distance sensing visible

At a glance: Make invisible distance sensing visible with a slinky pulse. About 80 min; 4 to 6 teams.

MATERIALS

- 4 to 6 metal slinkies
- floor tape
- station cards
- clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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- Redesign one variable, re-test, compare to baseline.
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DEBRIEF

- How is a slinky pulse like a sound wave?
- Why does timing an echo give distance?
- Where are ultrasonic sensors used?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Model and predictions	35
Station performance	25
Explanation	20
Team communication	20
Total	**100**

SOURCES IIHS SONAR Slinky

Pinhole Precision Challenge

The sharpest image, no lens

At a glance: Build the sharpest camera image with no lens. About 80 min; 4 to 6 teams.

MATERIALS

- cardstock
- foil
- tape
- pushpins or paper clips
- wax paper (through-view screen)
- light source

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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DEBRIEF

- Why is the image upside down?
- What did shrinking the hole do, and why?
- How is this related to your eye?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Image clarity	35
Concept explanation	25
Build quality	20
Redesign	10
Speed	10
Total	**100**

SOURCES Exploratorium Personal Pinhole Theater; NASA JPL Pinhole Camera

Low-Ropes Force Map Relay

Predict balance before the balance challenge proves you right

At a glance: Predict balance and stability before the balance challenge proves you right. About 80 min; 4 to 6 teams.

MATERIALS

- balance challenge cards (this packet)
- clipboards
- center-of-mass templates (this packet), a backless chair, painter tape, and a coin and small weight
- pencils

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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DEBRIEF

- When did your prediction match what you felt?
- How did lowering your center of mass help?
- Where else does balance physics matter?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct predictions	35
Force-map quality	25
Debrief explanation	20
Teamwork reflection	20
Total	**100**

SOURCES TeachEngineering center-of-mass and balance references

Hovercraft Hockey Hackathon

A puck that barely touches the floor

At a glance: Build a puck that moves by barely touching the floor. About 80 min; 4 to 6 teams.

MATERIALS

- recycled CDs or plastic discs
- pop-top bottle caps
- balloons
- hot glue station
- targets
- floor tape

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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DEBRIEF

- Why does the air cushion reduce friction?
- Why is more lift not always better?
- Where are air bearings used in real life?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Glide performance	30
Target competition score	25
Explanation	20
Redesign	15
Build quality	10
Total	**100**

SOURCES NASA JPL Hovering on a Cushion of Air; Science Buddies Hovercraft

INSTRUCTOR · PYS-10 · OPTICS + LIGHT

Spectra Sleuth Showdown

Two white lights, two hidden color fingerprints

At a glance: Can two white lights hide different color fingerprints? About 80 min; 4 to 6 teams.

MATERIALS

- diffraction glasses or gratings
- white LED and neon or gas sources
- incandescent source if available
- spectrum cards
- museum clue sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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DEBRIEF

- How can two white lights have different spectra?
- What does a line spectrum tell you?
- How do chemists make firework colors?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct matches	40
Spectrum sketches	20
Explanation	20
Exhibit connection	20
Total	**100**

SOURCES Science History Institute Flash Bang Boom; Exploratorium Spectra

BookBot Bin Logic Challenge

Retrieve by address, not by conveyor race

At a glance: Retrieve the right book by using bin addresses, not a conveyor race. About 80 min; 4 to 6 teams.

MATERIALS

- bin cards
- address labels
- order deck
- route mat
- timers
- optional grabber tool

FACILITATION ARC

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- Redesign one variable, re-test, compare to baseline.
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DEBRIEF

- Why store by address instead of by subject?
- What made one route faster than another?
- Where else is this storage logic used?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct retrievals	35
Low traffic conflicts	25
Routing or algorithm explanation	20
Efficiency	10
Teamwork	10
Total	**100**

SOURCES Temple Charles Library BookBot; Charles Library Circulating Collections

Accessibility Ramp Rescue Lab

A portable ramp that meets real constraints

At a glance: Design a portable ramp that works for a real user constraint. About 80 min; 4 to 6 teams.

MATERIALS

- foam board
- craft sticks
- dowels
- binder clips
- rulers
- free-rolling or pull-back car and a 200 g hanging weight set
- tape

FACILITATION ARC

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DEBRIEF

- Why does a gentler slope need more length?
- Which criterion was hardest to meet?
- Why design for accessibility from the start?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Load held	30
Slope meets target	30
Portability	20
Client criteria explanation	20
Total	**100**

SOURCES TeachEngineering Wheelchair Ramp Design

BACKUP ACTIVITIES

Ready alternates

Pencil Pressure Safety Lab

Spread a force before it dents the target

At a glance: Spread a force before it dents the target. About 60 min; 4 to 6 teams.

MATERIALS

- Pencils
- Foam
- Cardboard
- Weights
- Graph paper

FACILITATION ARC

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DEBRIEF

- Why does the same force do less damage over more area?
- Where is this used in safety gear?
- What made your spreader better?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Pressure spread	40
Explanation	25
Data quality	20
Redesign	15
Total	**100**

SOURCES IIHS Pencil Pressure

Domino Neuron Relay

Make a nerve signal survive a gap

At a glance: Make a nerve signal survive a gap. About 60 min; 4 to 6 teams.

MATERIALS

- Dominoes
- Rulers
- Tape
- Index cards

FACILITATION ARC

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DEBRIEF

- What happened when the gap got too wide?
- How is a bridge piece like a synapse?
- Why does myelin speed real signals?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Transmission success	35
Gap challenge	25
Explanation	20
Redesign	20
Total	**100**

SOURCES Exploratorium Domino Effect

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms

At a glance: Lift the load with smarter force, not stronger arms. About 60 min; 4 to 6 teams.

MATERIALS

- Classroom pulleys
- String
- Weights
- Spring scales

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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DEBRIEF

- Why did the movable pulley reduce effort?
- What did you trade for the easier lift?
- Where are pulley systems used in rescue work?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Setup accuracy	25
Explanation	20
Teamwork	20
Total	**100**

SOURCES TryEngineering Pulleys and Force

Barcode Checksum Rescue

Catch the fake barcode before the wrong part ships

At a glance: Catch the fake barcode before the warehouse ships the wrong part. About 60 min; 4 to 6 teams.

MATERIALS

- Printed barcode cards
- Dry erase boards
- Timers

FACILITATION ARC

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DEBRIEF

- How does one extra digit catch a typo?
- Why do false alarms matter in a warehouse?
- Where do you see check digits in daily life?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct detections	45
Low false alarms	20
Rule explanation	20
Teamwork	15
Total	**100**

SOURCES Computer science unplugged error-detection concepts